

Multi-Scale Deep Learning for Globally Applicable SCD Detection in Natural Hydrogen Exploration: Application to the Bristol Channel–Bray Fault System

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Abstract

Naturally produced hydrogen (H_2) has potential to be a scalable carbon-free energy resource, facilitating the urgent need to reduce reliance on fossil fuels. Sub-circular depressions (SCDs) are commonly associated with subsurface natural H_2 migration. Current deep learning approaches for SCD detection struggle to generalise across geographies and SCD morphologies. This study develops a new deep-learning model capable of detecting SCDs irrespective of their size or location. We subsequently utilise the model in southwest England to identify SCD clusters, and perform a soil agas sampling campaign, providing the first investigation into natural H_2 prospectivity in the United Kingdom. We employ a Mask R-CNN architecture using topographic data to mitigate the inconsistent spectral characteristics of SCDs. The model is trained using scale-normalised samples, and an image-pyramid is utilised at inference to maximise the receptive field of the model. The model achieves an F1 score of 0.83 in validation using a random test-train split, and a mean F1 score of 0.70 in cross-validation. At inference, the model is capable of detecting SCDs ranging from 20 to 2000 metres in diameter. This important contribution enables automatic cross-geography detection of SCD at a wider range of scales. In southwest England, 2752 SCDs are identified across 11460 km^2 , including the identification of an SCD cluster proximal to the Bristol Channel-Bray Fault system in Somerset. Here, SCDs are strongly spatially associated with mapped faults ($D^+=0.221$), especially the major Variscan strike-slip Watchet-Cothelstone-Hatch fault ($D^+=0.184$). Soil gas H_2 concentrations of up to 300 ppm from our targeted field campaign indicate the presence of an active H_2 migration system. The results demonstrate that the Bristol Channel-Bray Fault system has significant potential for further H_2 exploration in the United Kingdom, possibly correlating with well-documented H_2 discoveries near the Bray Fault, France.

1 Introduction

Natural hydrogen (H_2) has received significant attention for its transformative potential in the energy transition [1–3]. Hydrogen used in the energy sector is currently produced through expensive or carbon-intensive processes, such as steam-methane reforming or electrolysis [2]. Natural H_2 offers a carbon-free alternative to current costly and environmentally damaging production methods [3]. Only 1×10^5 Mt of naturally produced H_2 would provide a carbon-free resource yielding more than double the amount of energy within all known natural gas reserves [1]. Although current H_2 exploration techniques rely on soil gas surveys and legacy borehole data [3], recent discoveries of H_2 leaks associated with sub-circular depressions (SCDs) [4–8] have initiated an interest in deep learning based exploration strategies [9, 10].

Natural hydrogen is produced through subsurface geological processes, primarily serpentinisation or radiolysis [3, 11]. Hydrogen generated at depth can leak at the surface through structures known as sub-circular depressions (SCDs) or ‘fairy circles’. Their formation is poorly understood, but they have been proposed to form from mechanical compaction due to gas inflow and two-phase pore-pressure interactions [12]. Several clusters of SCDs associated with H_2 have been previously documented, commonly in sedimentary basins [4–8]. In Russia and the USA, SCDs occur in agricultural regions, displaying no spectral or vegetative anomalies [6, 7]. In contrast, SCDs found in Brazil and Australia are clearly identifiable through spectral data. They commonly exhibit a sparsely vegetated core surrounded by a densely vegetated rim [4, 5]. Semantic segmentation approaches utilising satellite data have been previously employed [9, 10], but these models struggle to generalise across terrain types and land-uses due to the distinct optical characteristics of SCDs depending on climatic and agricultural conditions. Similarly, SCDs can range from 20–2000 metres in diameter [4–6, 12], and current methods only utilise one resolution at inference, often leading to small depressions being undetected. We address these issues by developing a new globally-applicable instance segmentation model and inference pipeline capable of identifying SCDs of any size. We utilise the model in southern England, where favourable geological conditions related to the Bristol Channel–Bray Fault indicate potential for H_2 generation and migration.

The Bristol Channel–Bray Fault (BCBF) is a major Variscan strike-slip structure connecting the the Bristol Channel Fault Zone, England to the Bray Fault in Paris Basin, France. In southwest England, the BCBF bounds the Rhenohercynian Zone to the north and east, juxtaposing Devonian rocks from southwest England against the Avalonian terrane in central England. The Rhenohercynian Zone contains the ultramafic Lizard Complex, Cornwall, implying that ultramafic H_2 source rocks may exist at depth under the BCBF. The fault network associated with the BCBF could enable gas migration and H_2 leakage at the surface [13]. Several small SCDs (40–100 metres in diameter) have been previously documented along the Bray Fault, Paris Basin. Soil gas H_2 concentrations of up to 670 ppm have been measured in the region [14], while subsurface hydrogen concentrations in the Dogger Aquifer reach up to 52 volume percent [15]. Despite these discoveries in the Paris Basin, no investigations have been made into hydrogen prospectivity in the United Kingdom [16].

Here, we initially utilise a localised region-specific exploratory model to detect SCD clusters surrounding Variscan structures related to the Bristol Channel–Bray Fault system in the United Kingdom. The study develops and evaluates a novel cross-geography model and inference strategy capable of identifying SCDs at any scale in completely unseen regions, which can reproduce the predictions of the localised region-specific model in the UK. Finally, we report results from our soil gas sampling campaign at predicted SCD clusters in Somerset and provide the first investigation into H_2 prospectivity within the United Kingdom.

2 Methods

2.1 Data Preparation and Network Architecture

We utilise Digital Elevation Model (DEM) data to capture the morphological characteristics of SCDs without relying on geographically varying spectral signatures. The global 30

metre resolution Copernicus DEM is used as the primary data source [17]. Where available, higher resolution national datasets are used (e.g., 1 metre in UK, France, USA [18–20]). Orthorectified optical satellite data at 3 metre resolution is obtained from Planet Labs [21].

Five channels are input to the model: DEM, Slope, Laplace, Residual Relief (RR), and Hillshade (Figure 1a). The DEM provides overall topographic context, with slope and laplace channels explicitly carrying information on topographic variation and concavity. RR captures small-scale topographic variations by removing broader topographic trends, aiding the detection of small SCDs that are often masked by larger-scale topographic features such as slopes or valleys. Similarly, Hillshade records subtle terrain through artificial illumination. The chosen model architecture (Fig. 1b) is based on the Mask R-CNN [22]. This architecture is widely used in medical imaging for instance segmentation and object detection [23]. To adapt the Mask R-CNN for SCD detection, the anchor sizes are reduced to [20, 40, 62, 120, 220] pixels, and the anchor aspect ratios range is decreased to [0.75–1.25] to align with the sub-circular geometry of the depressions.

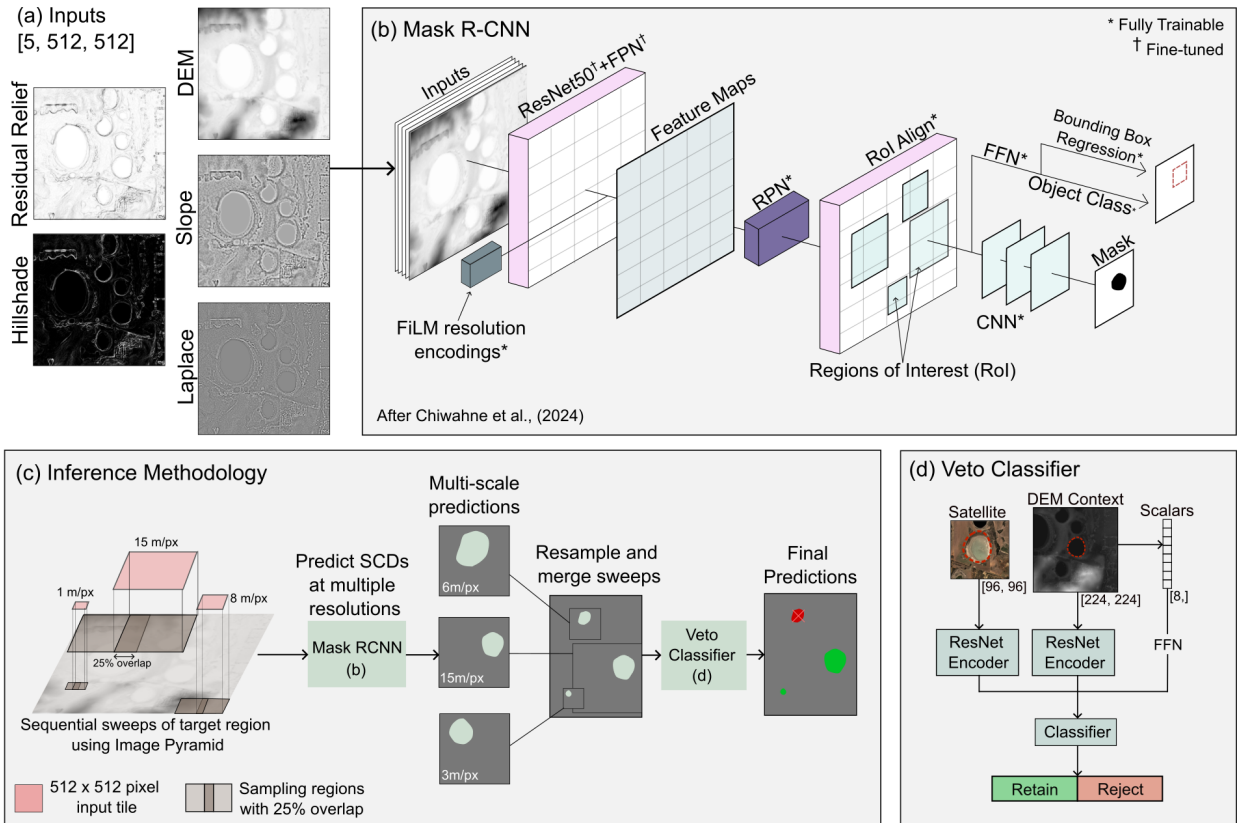


Figure 1: Schematic of deep learning methodology for scale-invariant SCD detection: (a) topography-based model inputs, (b) adapted Mask R-CNN architecture, (c) image-pyramid inference strategy with Mask R-CNN SCD proposals and a veto classifier, (d) framework of the veto classifier. FPN = Feature Pyramid Network, FFN = Feed Forward Network, CNN = Convolutional Neural Network, RPN = Region Proposal Network.

The training dataset contains 284 positive and 306 negative tiles. Negative tiles contain no SCD instances. Positive tiles contain at least one SCD instance and are generated from regions with previously identified SCDs: São Francisco Basin, Brazil (-16.452391N, -45.248511E [5]), Povorino, Russia (51.135297N, 41.911169E [6]), Moora, Australia (-30.619787N, 115.968694E [4]), Stiles, USA (31.472409N, -101.597082E), Coubert, France (48.672204N, 2.756956E). The training regions contain SCDs ranging in physical size and morphology to maximise the diversity of the dataset. A zoom or zoom-out operation is applied to each sample so that the target SCDs occupy 10–35% of the image area. The resulting resolution of each sample is encoded using Feature-Wise Linear Modulation (FiLM).

To detect SCDs at several scales, an image pyramid strategy is utilised at inference (Figure 1c). Predictions are generated at several scales by resampling the native DEM to a range of resolutions. A support-based voting system is used to merge the predictions from each resolution sweep. For a prediction to be retained, it must have been predicted at least two resolutions at >65% confidence, or one resolution at >90% confidence. The merged predictions are passed through a veto classifier, which either rejects or retains the prediction based on their spectral characteristics and broader DEM context (Figure 1d).

2.2 Transfer Learning and Training

The Mask R-CNN train set, consisting of 590 samples, is augmented using flips and rotations to increase the volume and diversity of the dataset. Samples are normalised independently, using Robust Scaling. DEM and Slope channels are also log-scaled. The Mask R-CNN is initialised with pretrained COCO weights and a stepped learning rate annealing scheme is used with different learning rates assigned to individual model blocks (Fig 1b, Table 1). The ResNet and FPN backbone is incrementally unfrozen at very low learning rates to enable fine-tuning to the task. The first layer of the backbone, Conv1, is trainable to allow the model to adapt to the new inputs.

Table 1: Learning rates used for each model component, in order of model architecture from left (first layer) to right (prediction heads).

Stage	Epochs	Conv1	ResNet	FPN	FiLM	RPN	RoI Heads
1	0–25	1×10^{-4}	Frozen	Frozen	1×10^{-3}	1×10^{-3}	1×10^{-3}
2	26–30	1×10^{-5}	Frozen	1×10^{-5}	1×10^{-3}	1×10^{-3}	1×10^{-3}
3	31–40	5×10^{-5}	1×10^{-6}	3×10^{-6}	1×10^{-4}	5×10^{-4}	5×10^{-4}
4	41–50	5×10^{-6}	5×10^{-7}	1×10^{-6}	1×10^{-5}	1×10^{-4}	1×10^{-4}

The veto classifier is trained on false positives outputted by the Mask R-CNN. For both models, an AdamW optimizer is used with a weight decay of $1e^{-4}$. The model weights from the epoch with the highest validation F1 score are saved and utilised at inference. The Mask R-CNN model is validated using Leave-One-Region-Out Cross-Validation (LORO-CV) to evaluate generalisation. Model performance is measured using object-level F1, Precision, and Recall (Equations 1-3) instead of pixel-level metrics as instance mask quality is less important than object detection in an exploration context. A true positive (TP) prediction must satisfy one of two criteria: (1) the centroid of the predicted SCD within a corresponding ground truth SCD mask, or (2) the predicted SCD mask and the ground truth SCD must overlap (Intersection over Union ≥ 0.5).

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives} + \text{False Positives (FP)}} \quad (1)$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives (FN)}} \quad (2)$$

$$\text{F1} = \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

2.3 Field Target Selection and Soil Gas Sampling

A region-specific UNET was developed to assist manual SCD identification in southwest England before the Mask R-CNN pipeline was developed due to scheduling constraints regarding fieldwork dates. The model was trained on 83 hand-picked depressions in the target region and used three input channels: DEM, Hillshade, and Topographic Position Index. The predictions of this model informed field target selection and to later evaluate inference predictions of the Mask R-CNN model.

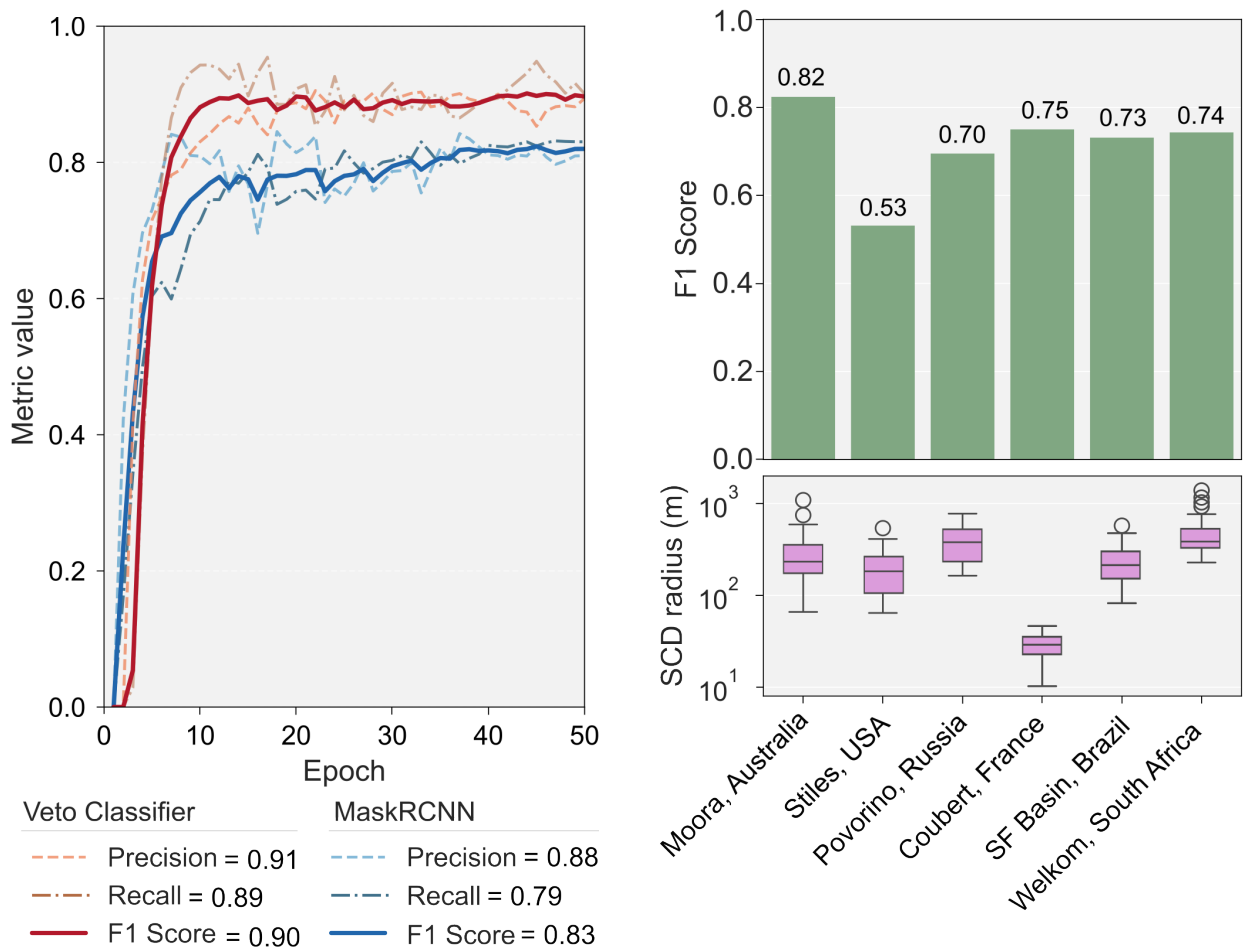
Sixty hours of fieldwork was undertaken in over seven days in June 2026. In Somerset, Soil gas measurements were taken at 15 SCDs, and a separate 21-kilometre coastal transect was performed over three days, with 34 sample sites at approximately 500 metre intervals. In Cornwall, a 10 kilometre transect was performed, and three potential SCDs were investigated. Soil gas sampling was performed using a Gembio portable gas analyser. The analyser tube was placed at 1 metres depth in the soil. The holes were drilled using percussion action only to limit the risk of friction-related H_2 artefacts, as first documented by Halas et al. [24]. The temperature of the drill bit was monitored to ensure that no rotary friction occurred. At each locality, at least four drill holes were sampled. Composition analysis of four gas samples collected in the field was performed by IsoLab. The upper background threshold of H_2 concentration is calculated using the Median Absolute Deviation (MAD). A one-sided

Kolmogorov-Smirnov (KS) D^+ statistic is used to analyse SCD enrichment near faults, with significance testing corroborated through Monte Carlo simulations.

3 Results

3.1 Model Performance

The Mask R-CNN successfully identifies SCDs in a random train-test split, achieving an F1 Score of 0.83, with a precision of 0.88 and recall of 0.79 (Fig. 2a). In LORO-CV, the model achieves an mean F1 score of 0.71, despite the unseen regions often containing different SCD morphologies and sizes (Fig. 2b). For example, the model achieves an F1 score of 0.75 on the validation fold for Coubert, France, where the median SCD radius is 29 metres. In Moora, Australia, where the median SCD radius is 233 metres, the model achieves an F1 score of 0.82. The veto classifier achieves an F1 Score of 0.90 in a random train-test split.



(a) Learning curves for the Mask R-CNN model and veto classifier. Learning curves are smoothed with exponentially weighted moving averages.

(b) SCD size distributions and cross-validation F1 scores per region for the Mask R-CNN. Model performs well in most regions despite a large range of physical SCD sizes.

Figure 2: Visualisation of Mask R-CNN and classifier training and validation metrics.

The image-pyramid inference strategy enables the model to detect SCDs at any scale without prior knowledge of the typical depression morphology in the target region (Fig. 3). Across different regions, the model correctly identified SCDs ranging from 25 metres to 1.5 kilometres in diameter. Inference predictions from the São Francisco Basin, Brazil, achieve an object-level F1 score of 0.84, with a precision of 90%. Visually, the model can accurately distinguish background from potential SCDs, but can struggle to confidently classify candidate depressions, leading to lower recall and some false positives. In Moora, Australia, the model consistently identifies large SCDs (500 metres – 1.5 kilometre diameter), but smaller SCDs are often missed due to resolution constraints. In regions where a 1 metre DEM is available (e.g., UK, France), small depressions (<100 metres) are consistently identified by the model. However, a consequence of the fine detail offered by these 1 metre DEMs is the overprediction of anthropogenic structures, such as farm ponds.

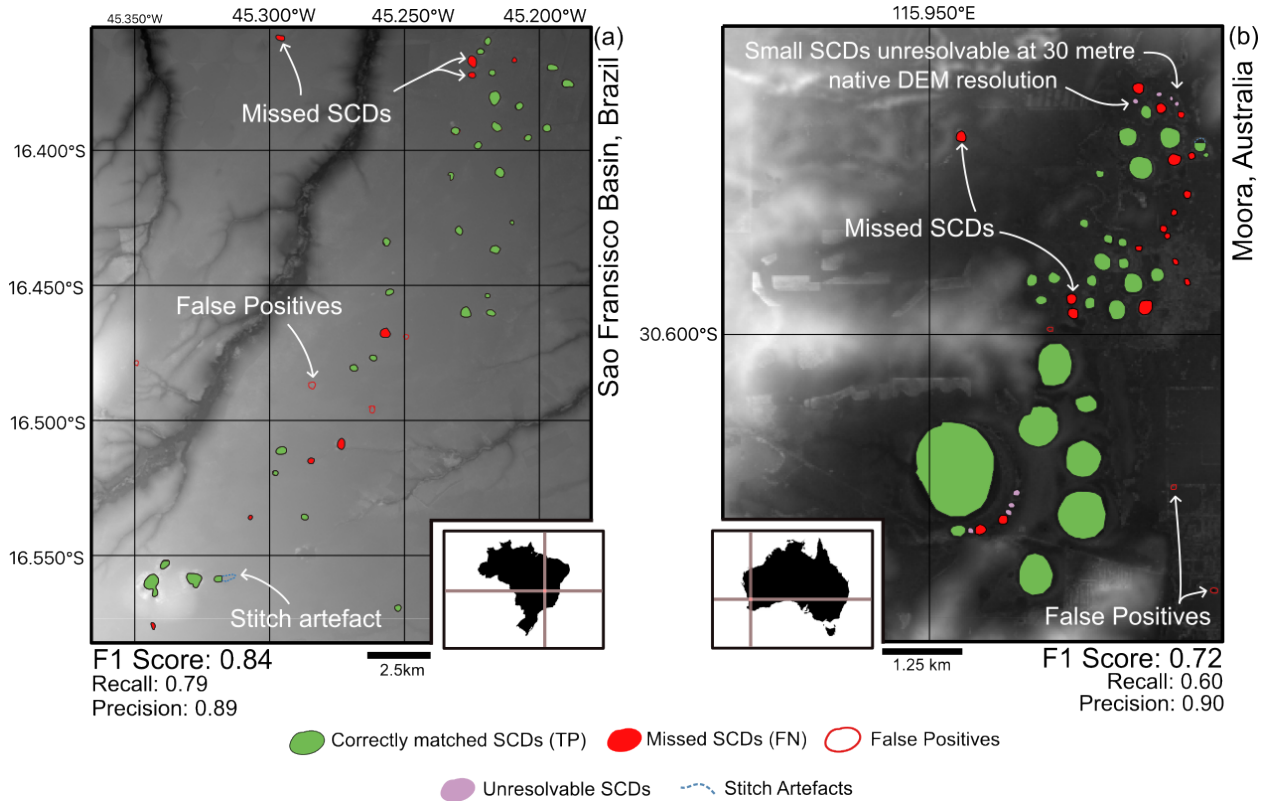


Figure 3: Inference predictions and object-based evaluation metrics: (a) predictions and ground truth comparison from the Sao Francisco Basin, Brazil, (b) predictions and ground truth comparison from Moora, Australia. Predictions are generated using an image pyramid with sweeps at 3, 6, 9, 12, and 15 metres per pixel.

At inference, the Mask R-CNN can generate hundreds of predictions across multiple resolution sweeps. The veto classifier can remove from 40–90% of these predictions depending on whether the region contains real SCDs. Despite the majority of the removals being correctly identified non-SCDs, the veto classifier occasionally removes valid SCD candidates. Similarly, false positive SCDs occasionally pass through the veto classifier, especially when the candidate structure is small (<100 metres in diameter) because the spectral resolution (3 metres per pixel) does not provide enough detail for a confident rejection. Artefacts from merging predictions from different resolutions can occasionally cause incomplete or blocky masks (e.g., stitch artefact in Figure 3a).

3.2 Model Inference and Cluster Identification in Southwest England

When deployed on 11,460 km² of southwest England, the model predicts 2752 SCD candidates (Fig. 4). Visual inspection of the predictions in this region reveals that the Mask R-CNN inference pipeline performs similarly well to the region-specific UNET developed for initial field target selection in the area, although it does predict more false positives than the initial model. The Mask R-CNN successfully identifies SCDs at the correct size range (20 to 50 metres in diameter). Most predictions were made at the appropriate scale level (1 metre per pixel), with very few large false positives predicted in coarser sweeps. However, due to the fine detail provided by the 1-metre DEM, the final predictions do contain anthropogenic false positives. The false positives are relatively evenly distributed and generally do not inhibit the capability of the model to identify SCD clusters (Fig. 4). The model does not distinguish between depressions based on origin; depressions associated with karst or bomb cratering and depressions associated with hydrogen leakage are predicted at the same confidence if they share similar morphologically characteristics.

Despite lower precision at inference compared to Mask R-CNN validation, the model detects several clusters of depressions within the prospected region. Two of the detected clusters are associated with karst: the Mendip Hills in the north and Dorset chalk dolines in the south [25, 26]. Outside of these clusters, the primary enrichment in SCD density occurs in the central area, around Taunton, in proximity to the Watchett-Cothelstone-Hatch Fault

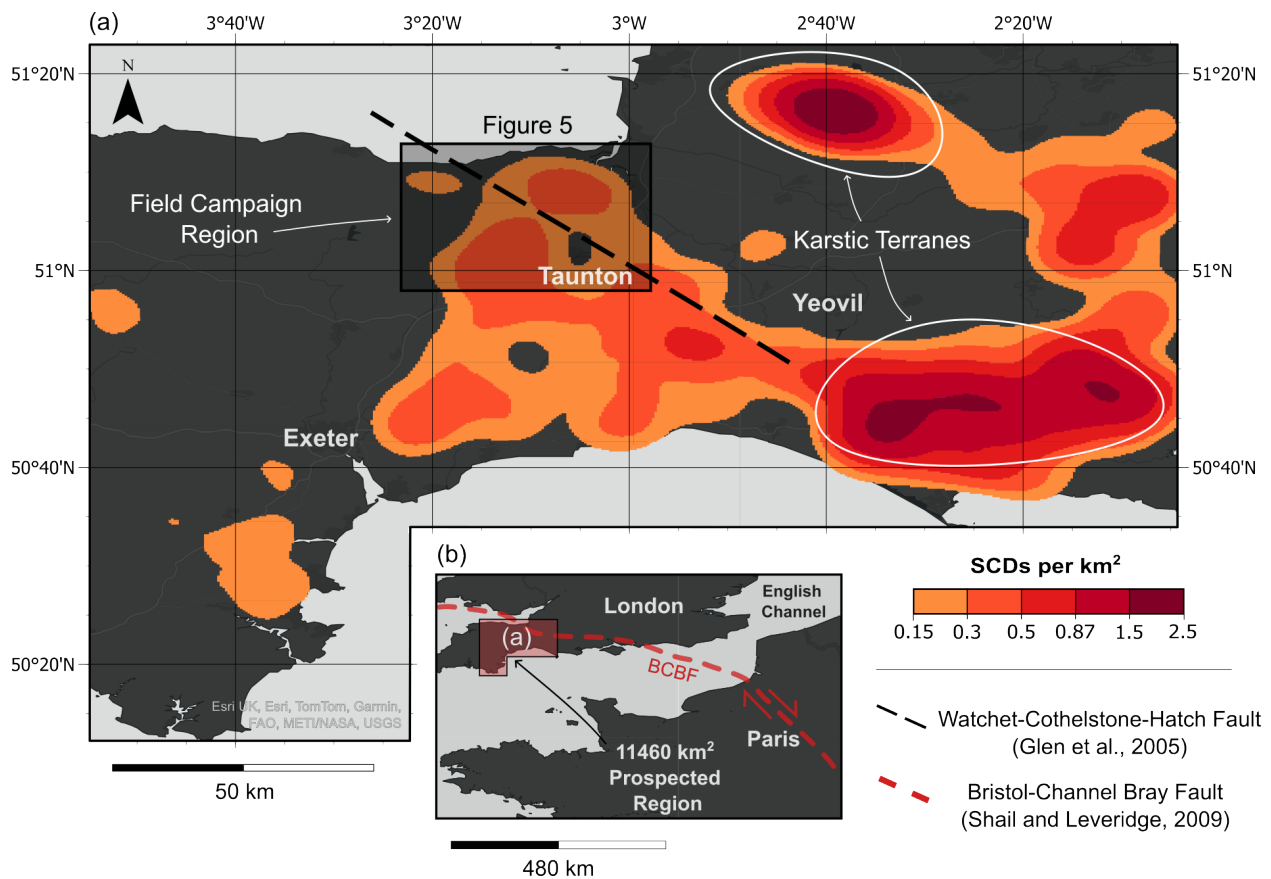


Figure 4: Model inference results in southwest England: (a) density of SCD candidates predicted using the Mask R-CNN in the 11460 square kilometre prospected region of southwest England. Predictions are generated using an image-pyramid with sweeps at 1, 3, 6, 9, 12, and 15 metres per pixel. (b) Location of the prospected region with respect to the major Bristol Channel–Bray Fault zone.

(WHCF). The field campaign region, on the southern margin of the Bristol Channel and proximal to the Bristol Channel–Bray Fault Zone, lies within this broad area of enrichment. The model detects 92 SCD candidates in this region, including the correct identification of 10 out of 15 SCDs visited in the field.

3.3 Field Results and Soil Gas H_2 Concentrations

Out of the 49 sample sites (34 along the transect, and 15 identified SCDs) in Somerset, 15 localities exhibit average H_2 concentrations exceeding the regional background upper threshold (9 ppm). A maximum concentration of 300 ppm is detected at Blue Anchor Beach. The median concentration of the 34 localities along the transect is 4.75 ppm. In total, the maximum concentrations at 9 localities measure above 40 ppm. Out of the 15 investigated SCDs, 7 exhibit mean soil gas concentrations above the background threshold, with four of these SCDs also exhibiting mean H_2 concentrations above 20 ppm. Five SCDs exhibit mean H_2 concentrations lower than 5 ppm. The SCDs in the region are small, with an average diameter of 27 metres and a depth of 1.8 metres. Due to the undulating topography of the region, the depressions are often not fully closed on the downslope side.

Localities associated with SCDs commonly contain higher soil gas H_2 concentrations compared to non-SCD localities, with mean concentrations of 11.6 and 7.2 ppm, respectively. However, the average hydrogen concentrations of SCD and non-SCD localities are not statistically separable into two populations (Mann–Whitney, $p = 0.247$). In comparison, the means of the peak H_2 concentrations measured at each SCD and non-SCD locality are 30.3 and 13.3 ppm, respectively. Using the maximum recorded values at each site, SCD and non-SCD localities are statistically separable (Mann–Whitney, $p = 0.0342$).

Hydrogen concentrations are highly variable both within and surrounding the SCDs (Figure 5b), demonstrating local heterogeneity in gas migration and accumulation within the soil. Depression rims generally exhibit higher H_2 concentrations than the depression cores, which commonly exhibit background values (approximately 0–5 ppm). These inactive cores bring

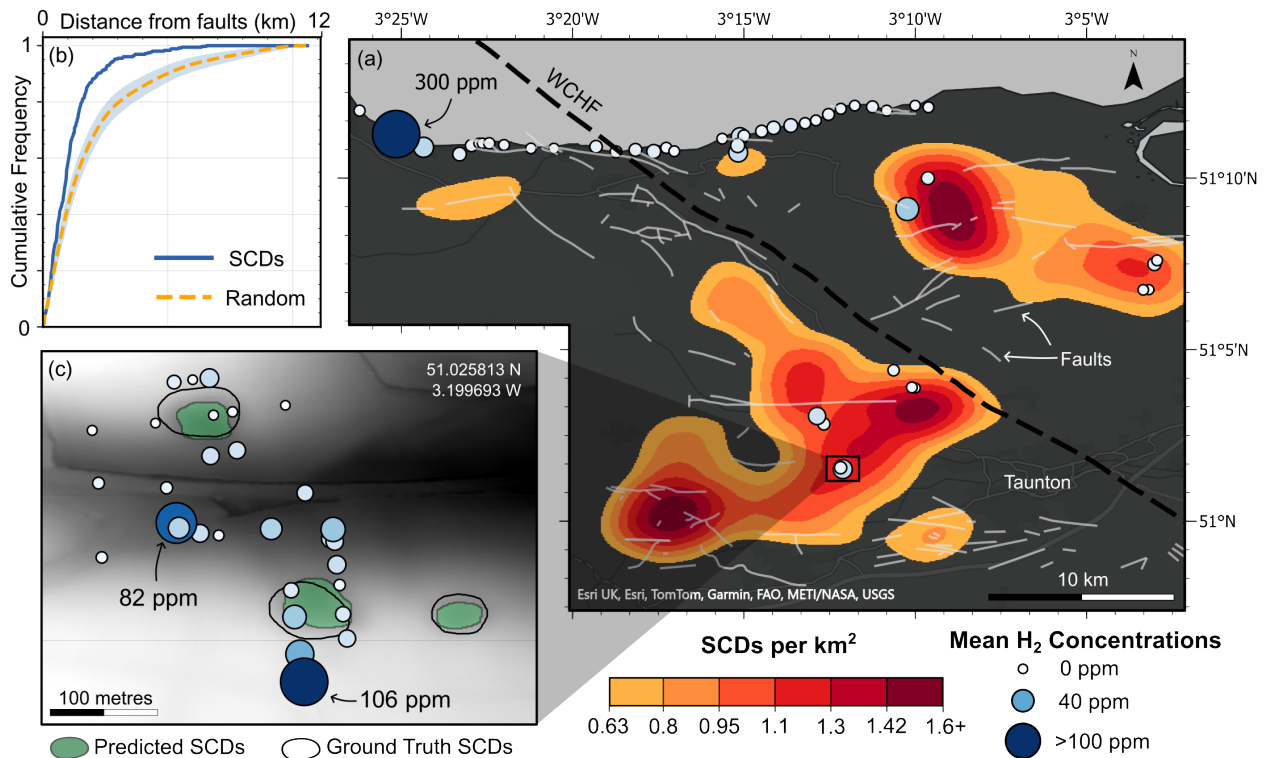


Figure 5: Model predictions and field campaign results in the Bristol Channel–Bray Fault Zone, Somerset: (a) soil gas H₂ measurements along a 21 kilometre coastal transect and 15 SCD localities, with model SCD prediction densities. Mapped faults [27] are indicated in white, (b) distance distribution analysis demonstrating a spatial association between predicted SCDs and mapped faults, (c) soil gas H₂ measurements at a small cluster of three predicted and visited SCDs near Oake Green. The cores of the depressions contain no H₂ anomalies, while the rims and surroundings exhibit up to 106 ppm.

down the average concentrations measured in SCDs; the median of the maximum concentrations measured at each SCD was 25 ppm. At every SCD, the highest H₂ concentration was detected either on the depression rim or 10–50 metres outside the structure.

There is no clear spatial correlation between elevated H₂ signals and the Watchett-Cothstone-Hatch Fault (WCHF), a major strike-slip Variscan fault. Similarly, there is no consistent relationship between visible coastal faults and increased hydrogen concentrations. However, spatial analysis and Monte Carlo simulations demonstrate highly significant concentration of predicted SCDs near to mapped faults (KS $D^+ = 0.221$, KS $p = 1 \times 10^{-16}$, Monte Carlo $p < 0.001$, 5000 simulations). A strong spatial correlation is retained when analysis is restricted to the Watchett-Cothstone-Hatch Fault (KS $D^+ = 0.184$, KS $p = 1 \times 10^{-11}$, Monte Carlo $p < 0.001$, 5000 simulations). Observed SCD enrichment compared to a random distribution is maximised at 6.3 kilometres from the WCHF, and 1.12 kilometres when all mapped faults are analysed.

Conversely, in the Lizard Complex, Cornwall, the model detects less than 5 SCDs and no appreciable soil gas H₂ concentration anomalies are observed. A maximum concentration of 50 ppm is detected at one locality. One SCD exhibits high CO₂ concentrations, although no H₂ is present.

4 Discussion

4.1 Model Performance

LORO–CV results demonstrate that the Mask R-CNN model can generalise across geographies and scales, allowing it to be deployed in completely unseen regions. Inference results in out-of-fold regions demonstrate robust geography and scale generalisation of the proposed model. Utilising a scale-normalised image-pyramid inherently largens the receptive field of the model, allowing identification of SCDs of different sizes. As discussed above, SCD sizes can differ by up to two orders of magnitude (Fig. 2b). Current models for SCD detection

only utilise one resolution, limiting the range of detectable SCDs to that of the receptive field of the underlying neural network. Although these models are more tailored to a specific resolution [9, 10], and may achieve better results in some regions, they are not applicable or generalisable to regions with smaller SCDs, such as the UK and France. Similarly, they are not applicable for unseen exploration in regions with SCDs that are not the same scale as the depressions in the respective training dataset. As such, a scale-normalised image-pyramid inference approach is a crucial improvement over other current SCD detection strategies for world-wide H_2 exploration in previously unstudied areas. These results are especially positive given the small dataset size (590 samples), and indicate the promising potential of the architecture if utilised with a larger dataset.

However, the model does not achieve full scale invariance, with smaller SCDs more commonly missed by the model despite the image-pyramid strategy. The failure mechanism here is clear; while spectral identification can utilise high-definition (3 metres per pixel) satellite imagery, topography-based identification relies on the global Copernicus 30 metre DEM unless a better national open-source dataset is available. In regions without a higher resolution DEM, SCDs smaller than approximately 60 metres in diameter are unresolvable, as they comprise only a few pixels at the native 30 metre per pixel resolution of the Copernicus DEM. This is primarily a data issue, and no image pyramid implementation can resolve resolution limitations of the source data. For example, in regions where a fine national DEM is available (e.g. 1 metre DEMs in the UK and France), small depressions are consistently identified by the model. While spectral-based models can achieve greater scale-invariance, the highly variable optical and spectral signatures of SCDs around the world limit the cross-geography generalisation. Although SCDs are often clearly identifiable through spectral anomalies and satellite imagery in arid regions [5, 8, 28], in temperate, agricultural regions such as France, Russia and the UK, they display almost no consistent spectral or optical anomalies [6]. Topography-based models do not suffer from a similar issue, as the morphological characteristics of SCDs are relatively comparable across regions.

Inference over large areas (several 1000s km^2) can cause lower precision compared to Mask R-CNN performance in validation due to large class imbalance between the normal background and rare SCDs. False positives are more common with a 1 metre DEM due to increased anthropogenic features and fine-detail, despite utilising negative training with these resolutions. For example, the model achieves an F1 score of 0.53 in USA, where the DEM resolution is 1 metre. Here, the structures can exhibit fluvial features, impacting model performance. Moreover, samples derived from fine DEMs are underrepresented in the training data due to data availability constraints. Coarser DEMs only reveal larger-scale topographic information and the model suffers less from overprediction problems. The model also predicts depressions that are not formed from H_2 migration, such as known karstic sinkholes. However, these depressions can provide a preferential pathway for gas flow, and are still targets for model detection. Further field investigation will be required to validate whether these karstic clusters identified by the model in southwest England are associated with H_2 or not.

4.2 Hydrogen Prospectivity in Southwest England

Soil gas H_2 concentrations measured in this study provide the first promising indication of an active H_2 migration system in the United Kingdom. The maximum concentration of 300 ppm measured at Blue Anchor Beach exceeds the both regional median (4.75 ppm) and previously documented typical background values (0–7 ppm [29]).

SCDs in the region are markedly smaller (20–50 metres in diameter) than structures identified in other regions of active H_2 migration (e.g., Russian Craton; [6]), and often present a unique unclosed morphology due to undulating terrain. Despite morphological differences, the SCDs often exhibit the typical elevation of H_2 concentrations around the depression rim compared to the core. This is a characteristic trait of topographical structures formed from gas-driven expansion and subsequent collapse, leaving a low-permeability core and a ring of preferential flow pathways around the depression rim [5–7, 12]. The presence of an ‘inactive’ subgroup of SCDs demonstrates that not all visited or predicted depressions are currently associated with H_2 migration, although some ‘inactive’ structures may originate from previous H_2 migration.

Statistically significant clustering of predicted SCDs near mapped faults strongly indicates fault-related gas migration. The governing WCHF demonstrates a similar statistical significance on the spatial distribution of SCDs, suggesting that it may exert a primary control on H_2 migration and subsequent SCD formation. The WCHF inherently exhibits a much broader control (peak enrichment distance = 6.3 km) compared to the effect of all mapped faults (peak enrichment distance = 1.12 km). The broader control of the WCHF implies that it may act as a primary migration pathway, feeding smaller faults that distribute (or previously distributed) H_2 throughout the region. Although present-day H_2 signals do not correlate with these faults SCD clusters may have formed around previous migration pathways which are now inactive.

Recent investigations into H_2 migration in the Paris Basin, France, have revealed soil gas H_2 concentrations of 220 ppm along the Bray Fault [30]. The geological link between these two regions (Bristol Channel Fault and Bray Fault) and their comparably high H_2 concentrations may (tentatively) imply the presence of a broader French–English H_2 system associated with Variscan structures such as the Bristol Channel–Bray Fault system. However, the source of H_2 in both regions is unclear. The ultramafic Lizard Complex in southwest England forms part of the Rhenohercynian Suture [15, 31]. Although we detected no H_2 anomalies in the ultramafic Lizard Complex, ultramafic rocks (serpentine, peridotite) readily facilitate the production of H_2 through serpentinisation at the correct temperatures (200–300 °C) and silica conditions [32]. The presence of ultramafic rock along the Rhenohercynian suture, which is bounded by the Bristol Channel–Bray Fault to the north and east [31], may indicate a possible ultramafic source rock at depth. Moreover, the major Variscan faults may extend down to the upper mantle and provide a direct hydrological link to basement ultramafics, facilitating large-scale H_2 production and migration. Magnetic and gravimetric data indicate the presence of granite in the north of the Paris Basin, indicating the possible role of radiolysis in H_2 generation [15].

The lack of H_2 observed in the Lizard Complex contradicts the British Geological Survey review of natural hydrogen potential in the United Kingdom, which rates the area with higher prospective grade than the Somerset region [16]. Although the Lizard Complex comprises ultramafic rocks, the temperatures and pressure are too low at the surface for H_2 generation to occur. Given the anomalous concentrations found in Somerset, our results indicate that hydrogen prospectivity studies and exploration require automatic SCD detection to inform on any surface H_2 leakage, and characterising the lithology and terrane type is insufficient, as source rocks are likely buried at depth.

The measured anomalous concentrations in Somerset (50–300 ppm) are appreciable but moderate in a global context (e.g., thousands of ppm; [5–7]). However, in the Paris Basin, on the southwestern side of the Bristol Channel–Bray Fault system, only 9 out of 240 measurements exceed 100 ppm, despite the underlying aquifer containing up to 52 volume percent H_2 [14, 15]. As such, the measured H_2 concentrations and significant fault-related SCD clustering demonstrate substantial potential for further investigation into H_2 migration in southwest England. Further isotopic investigation of collected H_2 gas is needed to confirm the link between H_2 discoveries in the Paris Basin and Somerset. Similarly, geophysical investigation (i.e. seismic, gravimetric, magnetic) into the subsurface is required to understand migration pathways and identify potential granitic or ultramafic source rocks.

5 Conclusion

The performance of the model in validation and inference confirms that the Mask R-CNN image–pyramid pipeline provides a flexible, cross-geography tool applicable for automatic SCD detection. Similarly, utilisation of topographic data removes the limitation associated with the inconsistent spectral characteristics of SCDs. These contributions provide a crucial improvement to first pass natural hydrogen exploration, and specifically enable the first automatic identification of SCDs in the United Kingdom. Several clusters of depressions are identified in Somerset, on the southern margin of the Bristol Channel. Significant enrichment of SCDs to mapped faults in the area, specifically the major Variscan Watchet-Cothelstone-Hatch Fault, indicate localised H_2 migration throughout the fault network. Soil gas concentrations of up to 300 ppm further demonstrate the promising potential of the region for

further H₂ exploration, especially considering the geological link with highly documented discoveries in the Paris Basin.

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AI has not been used at any point in the generation of text in this paper. OpenAI’s ChatGPT 5.6 Sol and Anthropic’s Claude Sonnet 5 was used for assistance with modelling and code generation, as outlined in the GitHub repository associated with this study. These models were used throughout the code development process for debugging. Any code that was generated by AI was reviewed, modified and manually integrated into the code base. In the repository, an acknowledgement to AI can be found wherever these models influenced the code. The submissions reflects my own work, methodology ideas and implement approach as declared in the Academic Integrity Declaration. Github Copilot Free was used to assist code documentation. I would specifically like to clarify that AI was NOT used in the production of any figures. ChatGPT was used to create the L^AT_EX code for Table 1.

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